

Are Harris County Drivers the Worst in Texas?

An Editable Beamer Presentation

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Are Harris County Drivers the Worst in Texas?

Senior project comparing selected Texas counties using crash rates, confidence intervals, and regression.

Selected counties: Harris, Travis, Bexar, Montgomery, Hays, Medina, Hidalgo, Hale, and Lipscomb.

Research Question

Main Question

Is Harris County the county with the worst drivers among the counties in this sample?

Working Idea

This project treats “worst drivers” as a question of:

- crash frequency relative to population
- fatal crash risk
- risky driving patterns such as speeding, DUI, and distraction

Why Total Crashes Are Not Enough

A large county will usually have more crashes simply because it has more people and more traffic.

That means total crashes alone can be misleading.

Better comparison measures

- crash rate per 100,000 people
- fatal crash rate per 100,000 people
- proportions involving speeding, DUI, and distracted driving

Counties in the Study

Table 1: Counties included in the analysis

County
Harris
Travis
Bexar
Montgomery
Hays
Medina
Hidalgo
Hale
Lipscomb

Creating New Variables

To compare counties fairly, several derived variables were created.

$$\text{Crash Rate} = \frac{\text{Total Crashes}}{\text{Population}} \times 100,000$$

$$\text{Fatal Crash Rate} = \frac{\text{Fatal Crashes}}{\text{Population}} \times 100,000$$

$$X = \text{Speed Percent} = \frac{\text{Speed Crashes}}{\text{Total Crashes}}$$

$$Y = \text{DUI Percent} = \frac{\text{DUI Crashes}}{\text{Total Crashes}}$$

$$Z = \text{Distracted Percent} = \frac{\text{Distracted Crashes}}{\text{Total Crashes}}$$

Summary of Derived Variables

Table 2: Derived Variables

County	Crash_Rate	Fatal_Rate	SpeedPercent	DUIPercent	DistractedPercent
Harris	2301.668	10.912	0.324	0.029	0.086
Travis	1137.039	9.958	0.192	0.074	0.323
Bexar	2238.547	9.458	0.155	0.034	0.366
Montgomery	1592.321	7.606	0.370	0.039	0.146
Hays	1221.680	8.818	0.311	0.067	0.253
Medina	1475.806	19.419	0.234	0.055	0.228
Hidalgo	1788.161	6.140	0.336	0.045	0.095
Hale	1437.017	12.578	0.274	0.055	0.236
Lipscomb	910.747	36.430	0.200	0.000	0.080

Ranking Counties by Crash Rate

Table 3: Counties ranked by crash rate per 100,000

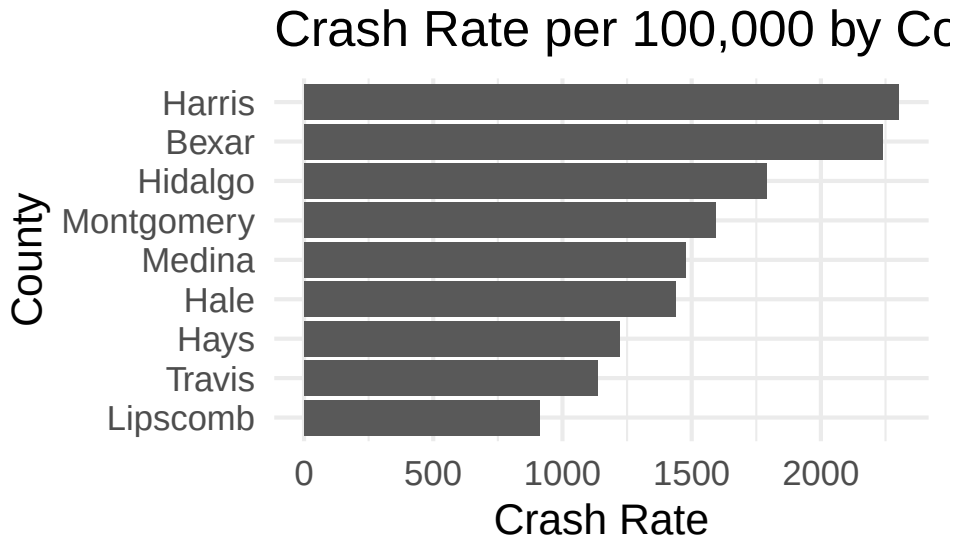
County	Crash_Rate	Total_Crashes	Population
Harris	2301.67	115173	5003892
Bexar	2238.55	48522	2167567
Hidalgo	1788.16	16601	928384
Montgomery	1592.32	12352	775723
Medina	1475.81	836	56647
Hale	1437.02	457	31802
Hays	1221.68	3602	294840
Travis	1137.04	15872	1395906
Lipscomb	910.75	25	2745

Crash Rate per 100,000 People

- Harris County: 2301.7 (highest)
- Bexar County: 2238.5 (very close)
- Travis County: 1137 (lowest among large counties)
- Lipscomb County: 910.7 (lowest overall)

Harris has the highest crash rate, but only slightly higher than Bexar.

Crash Rate by County



Ranking Counties by Fatal Crash Rate

Table 4: Counties ranked by fatal crash rate per 100,000

County	Fatal_Rate	Fatal_Crashes	Population
Lipscomb	36.43	1	2745
Medina	19.42	11	56647
Hale	12.58	4	31802
Harris	10.91	546	5003892
Travis	9.96	139	1395906
Bexar	9.46	205	2167567
Hays	8.82	26	294840
Montgomery	7.61	59	775723
Hidalgo	6.14	57	928384

Brief interpretation

Fatal crash rate focuses on severity rather than frequency, so it can highlight counties where

Risk Factor Table

Table 5: Risk-factor proportions by county

County	SpeedPercent	DUIPercent	DistractedPercent	SeverityPercent
Montgomery	0.3705	0.0393	0.1459	0.0291
Hidalgo	0.3359	0.0453	0.0947	0.0211
Harris	0.3240	0.0291	0.0855	0.2503
Hays	0.3115	0.0666	0.2526	0.0455
Hale	0.2735	0.0547	0.2363	0.0416
Medina	0.2344	0.0550	0.2285	0.0514
Lipscomb	0.2000	0.0000	0.0800	0.0400
Travis	0.1920	0.0745	0.3235	0.0410
Bexar	0.1549	0.0341	0.3660	0.0202

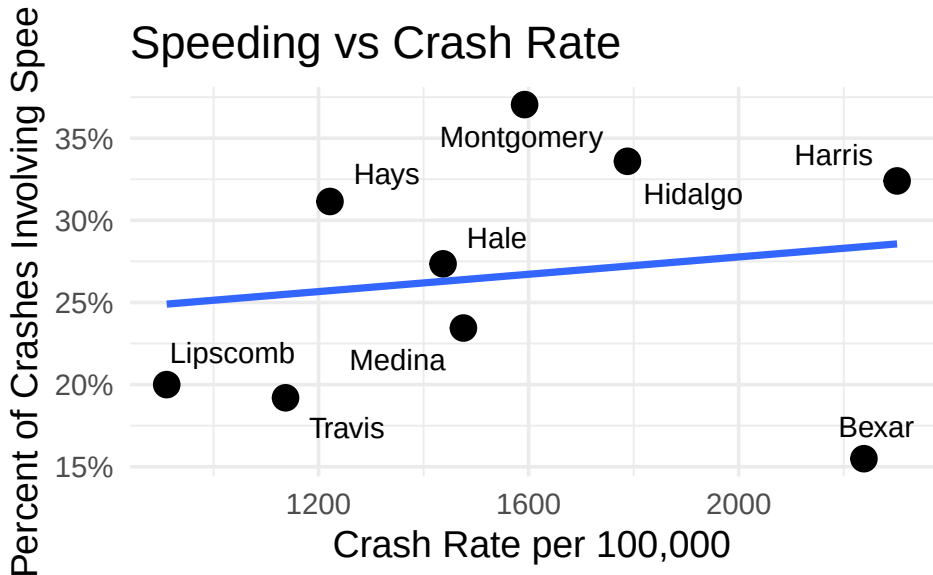
Driving Behavior Across Counties

- Speed-related crashes:
 - moderate across most counties
- DUI crashes:
 - relatively low percentages
- Distracted driving:
 - high in multiple counties, especially urban counties

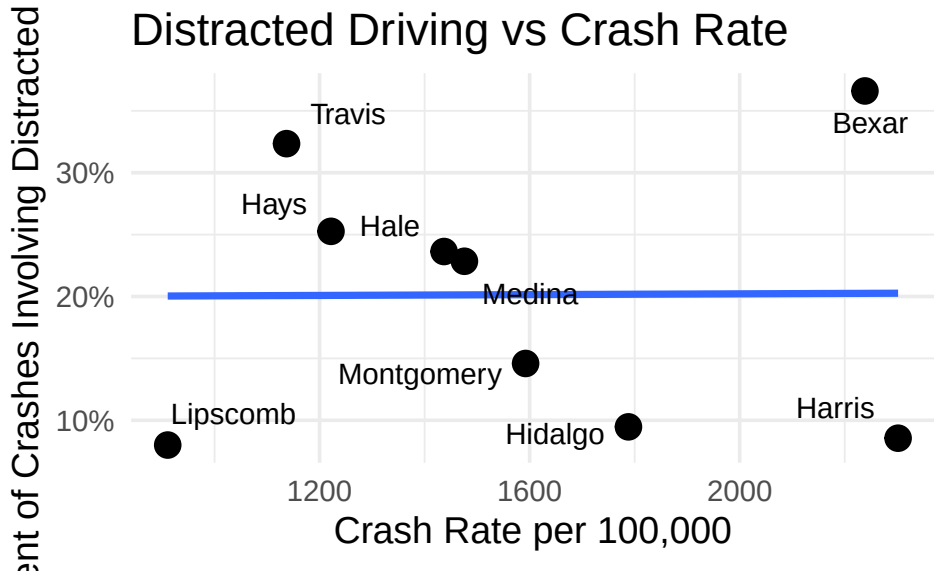
Insight

Risk behaviors vary, but no single factor dominates.

Speeding vs Crash Rate

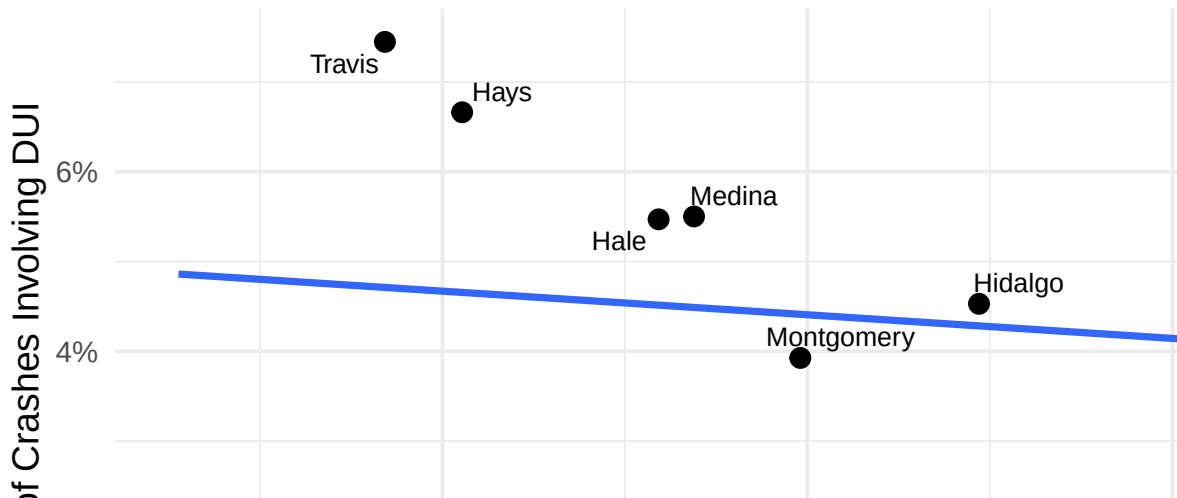


Distracted Driving vs Crash Rate



DUI vs Crash Rate

DUI Driving vs Crash Rate



Confidence Intervals for County Crash Rates

To estimate the true crash rate for each county, I used a 95% confidence interval.

$$\hat{p} = \frac{\text{Total Crashes}}{\text{Population}}$$

$$\hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where:

- $\hat{p}$ = observed crash proportion
- n = county population
- 1.96 gives a 95% confidence level

Purpose

This helps show whether differences in county crash rates are likely to be meaningful or just due to random variation.

95% Confidence Intervals by County

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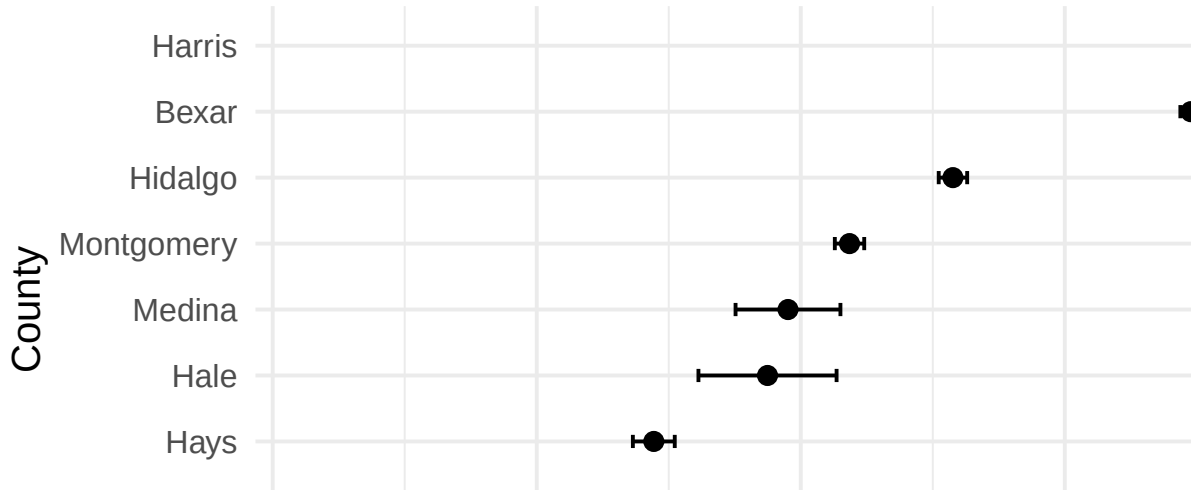
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\caption{Crash rates and 95% confidence intervals per 100,000 residents}
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County	Crash_Rate	Lower_CI	Upper_CI
Harris	2301.67	2288.53	2314.81
Travis	1137.04	1119.45	1154.63
Bexar	2238.55	2218.85	2258.24
Montgomery	1592.32	1564.46	1620.18
Hays	1221.68	1182.03	1261.33
Medina	1475.81	1376.51	1575.11
Hidalgo	1788.16	1761.20	1815.12
Hale	1437.02	1306.21	1567.82
Lipscomb	910.75	555.36	1266.13

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Crash Rates with 95% Confidence Intervals

Crash Rates with 95% Confidence Intervals



Interpreting the Confidence Intervals

- If two counties have intervals that do not overlap, that suggests stronger evidence that their crash rates are different
- If two counties have intervals that overlap a lot, then the difference may not be statistically meaningful

Harris vs Montgomery

- Harris County has a higher observed crash rate than Montgomery County
- But if their intervals overlap, then the data does not clearly show that Harris is worse
- This matters because neighboring counties may share highways, commuter traffic, and similar driving environments

Main Point

Confidence intervals help move the project beyond simple ranking and toward statistical comparison.

Regression Model

$$\text{Crash Rate} = \beta_0 + \beta_1 X + \beta_2 Y + \beta_3 Z + \varepsilon$$

- Goal: identify which behaviors increase crash rates

Are the results significant?

Table 6: Regression coefficients

term	estimate	std.error	statistic	p.value
(Intercept)	84.007	1372.552	0.0612	0.9536
SpeedPercent	5107.916	4239.003	1.2050	0.2821
DUIPercent	-16501.027	14148.702	-1.1663	0.2961
DistractedPercent	4237.233	3694.462	1.1469	0.3033

- All p-values > 0.05
- None of the variables are statistically significant

Interpretation

We cannot confidently say any factor causes higher crash rates.

Regression Findings

Table 7: Regression model fit

r.squared	adj.r.squared	sigma	statistic	p.value
0.25	-0.2	519.6527	0.5556	0.6667

- $R^2 = 0.25$
- Model explains only about 25% of the variation

Coefficients

- Speed → positive effect
- Distracted → positive effect
- DUI → unclear or negative effect

Interpretation of Results

Crash rates are not fully explained by:

- Speed
- DUI
- Distracted driving

Likely Explanation

Other factors may matter more:

- Traffic density
- Urban congestion
- Road design

Limitations

- Small sample size ($n = 9$ counties)
- Limited variables
- Regression results are not statistically significant

Implication

The results suggest trends, but not definitive causation.

Final Conclusion

- Harris County has the highest crash rate
- But:
 - only slightly higher than Bexar
 - It is not strongly explained by driver behavior

Final Answer

Harris does not clearly have the “worst drivers.”